At line #6, we choose X's left child, denoted with L in the figure, as its highest-priority child, and so we perform a right rotation (line #7), which produces the tree shown in figure 3.10.B.

Here in the figure we changed the priority of the node being pushed down to $+\infty$[5], but in the code we don't actually need to do that, we can just push down the node without checking priorities, until it becomes a leaf.

At this point X is not yet a leaf, although it just has one child, its (also former) right child R; therefore, we will enter another iteration of the while loop, and this time perform a left rotation, producing the tree shown in figure 3.10C; one more left rotation, and X finally becomes a leaf.

At this point we exit the while loop, and at line #10 we are sure that node is not the root (otherwise we would have caught this case at line #4), so it will have a non-null parent: we still need to disconnect the node from the tree by removing the pointer from its parent, and to do so we need to check whether it was a left or right child.

Once the right link has been set to null, we are done and the key was successfully removed.